

Warehouse robot coordination

A* search for a safe delivery wave

Fundamentals of AI

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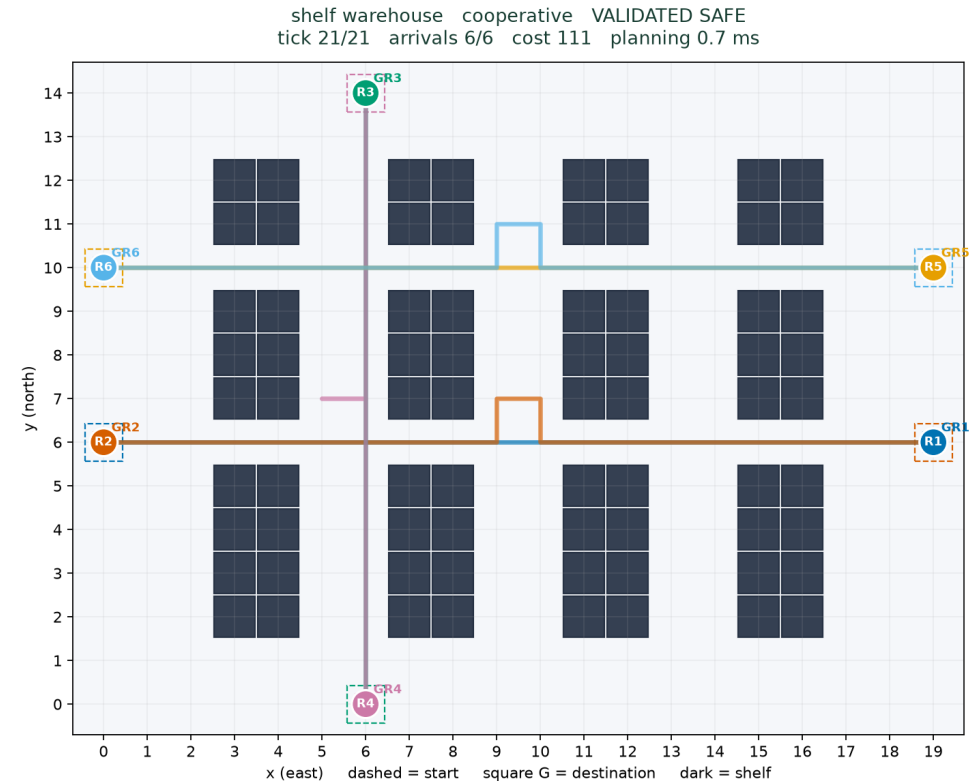
The warehouse problem

20 x 15 cells and six robots

Assigned starts and unique destinations

Crossing aisles connect shelf blocks

Safe completion is the primary objective



State actions and collision rules

Model	Rule
Coordinates	(x,y), east +x, north +y
Actions	N, S, E, W or wait, unit cost
Vertex conflict	Same cell at the same tick
Edge swap	Opposite traversal in one interval
Goals	Hold accepted terminal goals forever
Following	Allowed after another robot leaves

Spatial A* search

$$f(n) = g(n) + h(n)$$

$$\text{Manhattan } h = |x - \text{goal}_x| + |y - \text{goal}_y|$$

Cell toward (2,1)	g	h	f
(0,0)	0	3	3
(1,0)	1	2	3
(2,0)	2	1	3

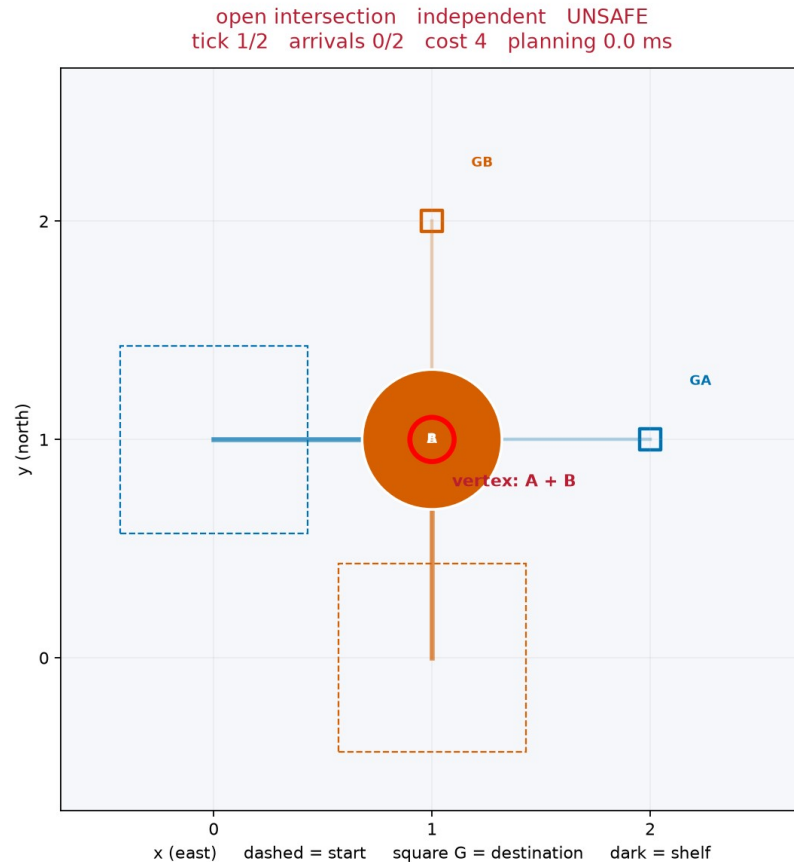
heapq frontier

Deterministic ties

Stale-entry check

$h = 0$ comparison

Independent routes can collide



Tick 1 shares the center

Both individual routes are shortest

Joint replay detects a vertex conflict

The saved plan is labeled UNSAFE

Space-time A* includes waiting

State (x,y,t)

Move or wait advances to t+1

Reservations constrain successors

Goal acceptance checks future occupancy

Tick	A	B
0	(0, 1)	(1, 0)
1	(1, 1)	(1, 0)
2	(2, 1)	(1, 1)
3	(2, 1)	(1, 2)

B waits one tick before the crossing

Reservation table semantics

Vertex $(cell, t)$ includes tick zero

Edge (u, v, t) represents interval t to $t+1$

Reject the reserved reverse traversal (v, u, t)

Permanent goal reservations continue indefinitely

Future finite visits also prevent early terminal acceptance

Implementation and planning flow

Scenario validation and spatial preflight

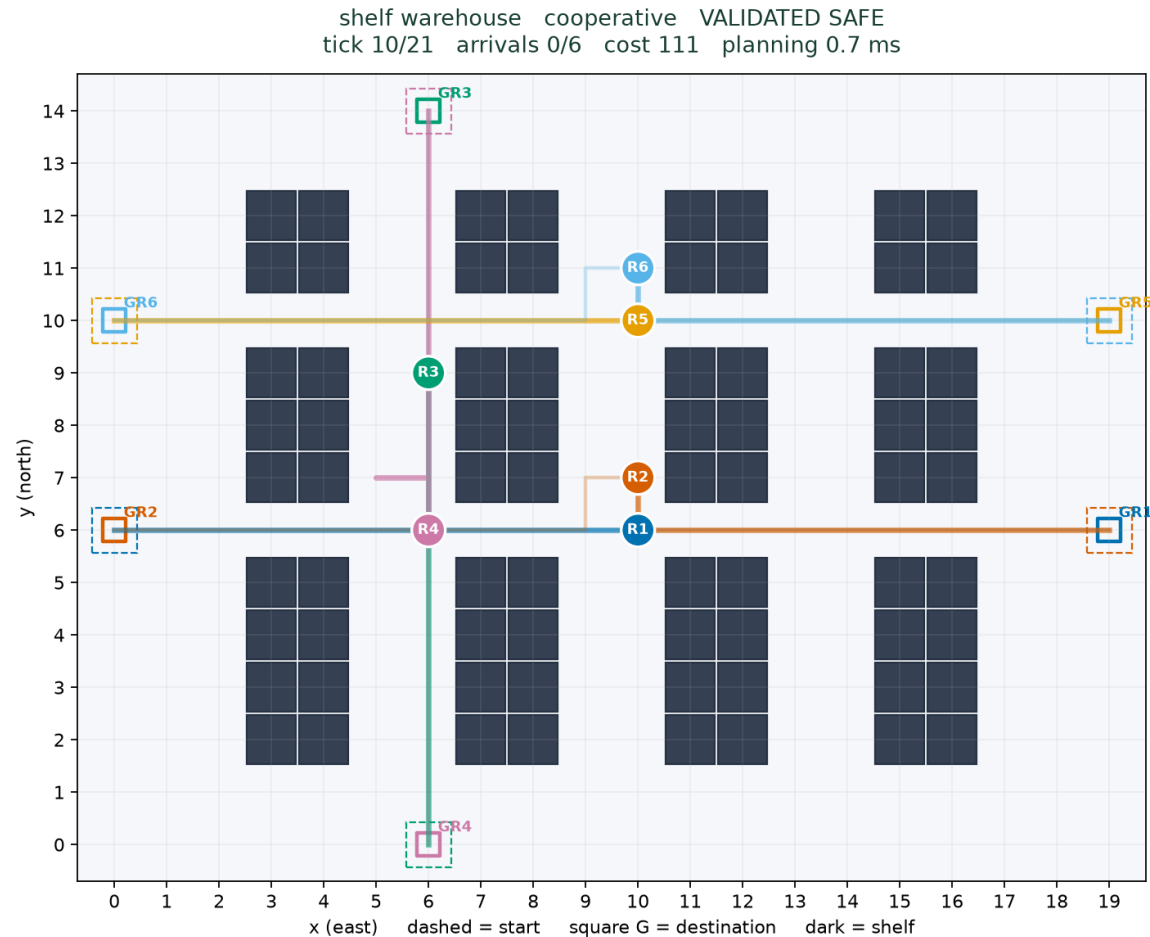
Fresh reservation table for each priority attempt

Space-time A* commits one route at a time

Independent joint validator checks complete paths

Self-contained plan JSON feeds replay and export

Simulation and live demo



Play/pause, Step and Reset

Playback speed and overlays

Cost 111, makespan 21

110 moves and 1 wait

Zero detected conflicts

Experiment design and fair comparison

Same 20 aggregate instances for all algorithms

Robot counts 2, 4, 6, 8 with five seeds each

Six separate diagnostic scenarios

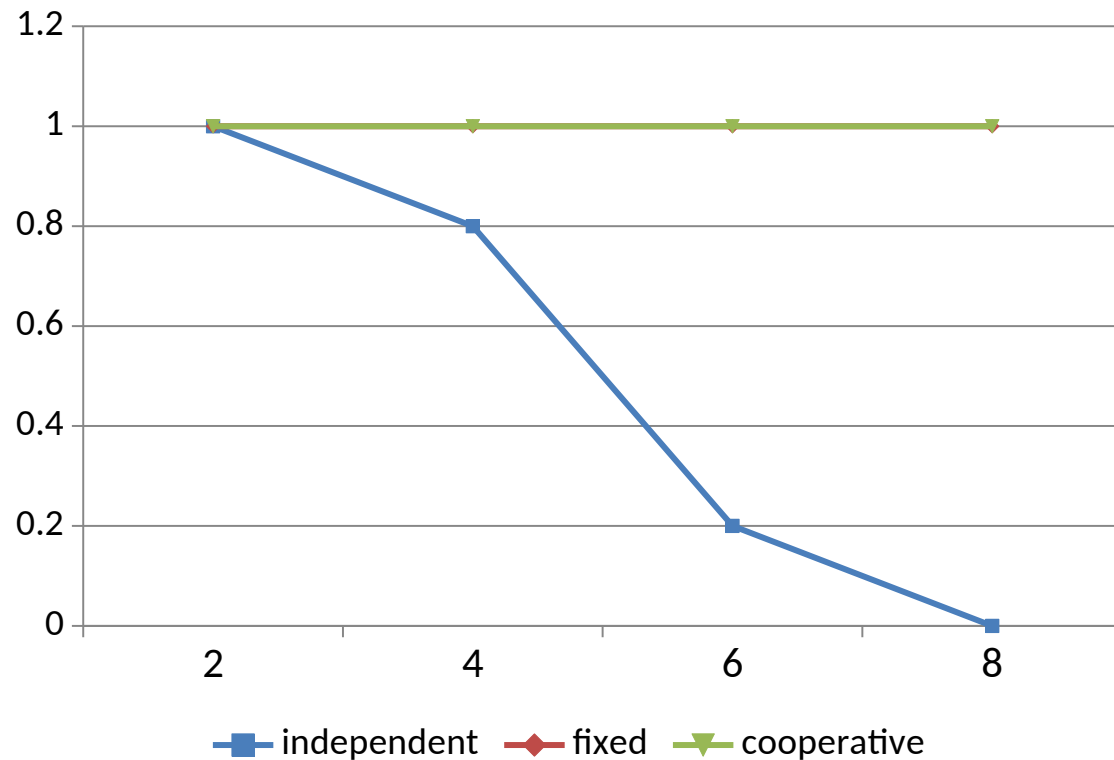
Horizons 40 and 80, expansion cap 30,000

Rates use all cases, costs use valid cases

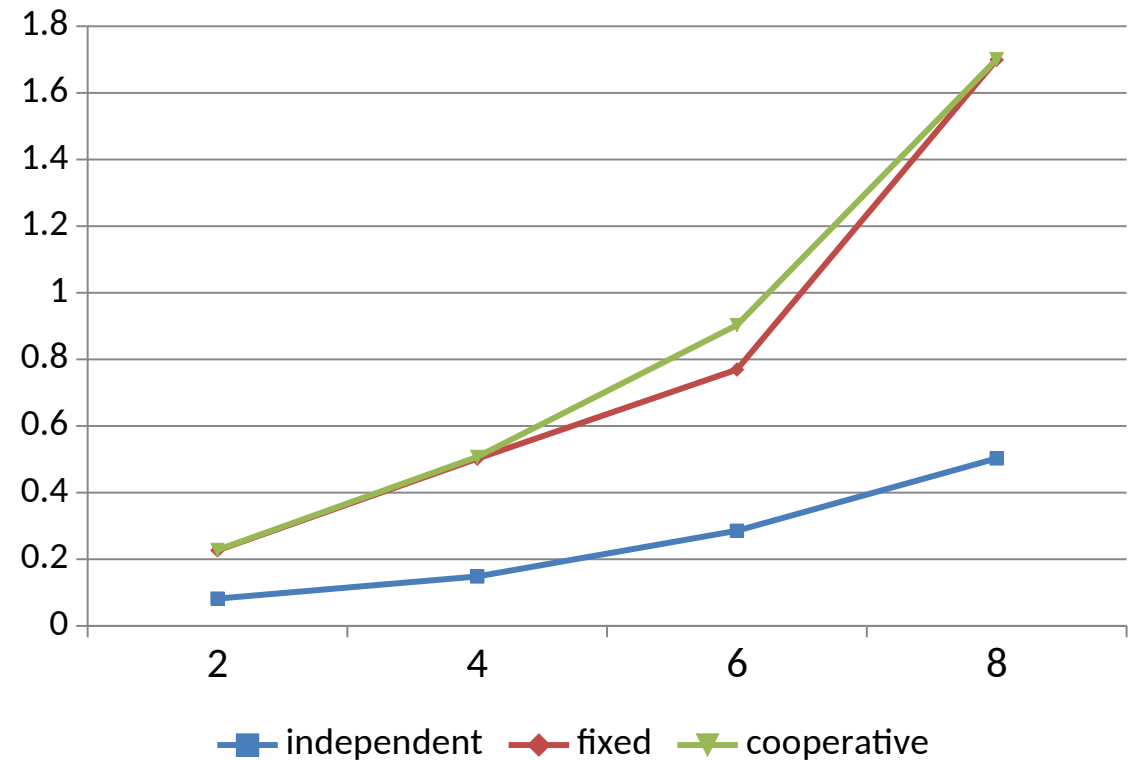
Timing includes failed attempts and excludes rendering

Measured completion and planning time

Safe completion rate



Mean planning time (ms)



Five seeds per robot count. Both coordinators safely complete 20/20 sampled cases.

Priority portfolios can increase travel cost

Robots	Fixed cost	Portfolio cost	Paired delta
2	24.6	24.6	0
4	44.6	44.6	0
6	75.4	77.8	2.4
8	114.2	114.2	0

Five common-success pairs per count

At six robots, higher mean cost accompanies a lower mean makespan

The solver returns the first valid priority attempt

Tests safety validation and failures

22 passed in 1.03s

50 seeded maps checked against an independent BFS oracle

Vertex, swap, following and permanent-goal tests

Order-sensitive retry and bounded impossible corridor

CLI, JSON reload, PNG and GIF export checks

Limits and future work

Prioritized planning is incomplete and order-dependent

The coordinator does not globally optimize cost or makespan

Finite horizons and expansion limits can reject feasible cases

Static shelves, point robots and centralized offline planning

Future work: tiny-instance CBS and physical movement constraints

Demonstrated findings and references

Default six-robot wave safely finishes in 21 ticks

Both coordinators solve all 20 aggregate samples safely

Independent validation protects every complete returned plan

Hart, Nilsson and Raphael (1968), DOI 10.1109/TSSC.1968.300136

Silver (2005), DOI 10.1609/aiide.v1i1.18726

Stern et al. (2019), arXiv:1906.08291